

CPU Scheduling Problems

1. Suppose that the following processes arrive for execution at the times indicated. Each process will run for the amount of time listed.

Process	Arrival Time	Burst Time
P ₁	0.0	8
P ₂	0.4	4
P ₃	1.0	1

- a. What is the average turnaround time for these processes with the FCFS scheduling algorithm?
 - b. What is the average turnaround time for these processes with the SJF scheduling algorithm?
2. Consider the following set of processes, with the length of the CPU burst time given in milliseconds:

Process	Burst Time	Priority
P ₁	2	2
P ₂	1	1
P ₃	8	4
P ₄	4	2
P ₅	5	3

The processes are assumed to have arrived in the order P₁, P₂, P₃, P₄, P₅, all at time 0.

- a. Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a larger priority number implies a higher priority), and RR (quantum = 2).
 - b. What is the turnaround time of each process for each of the scheduling algorithms in

part a?

- c. What is the waiting time of each process for each of these scheduling algorithms?
 - d. Which of the algorithms results in the minimum average waiting time (over all processes)?
3. The following processes are being scheduled using a preemptive, round-robin scheduling algorithm.

Process	Priority	Burst	Arrival
P ₁	40	20	0
P ₂	30	25	25
P ₃	30	25	30
P ₄	35	15	60
P ₅	5	10	100
P ₆	10	10	105

Each process is assigned a numerical priority, with a higher number indicating a higher relative priority. In addition to the processes listed above, the system also has an idle task (which consumes no CPU resources and is identified as P_{idle}). This task has priority 0 and is scheduled whenever the system has no other available processes to run. The length of a time quantum is 10 units. If a process is preempted by a higher-priority process, the preempted process is placed at the end of the queue.

- a. Show the scheduling order of the processes using a Gantt chart.
 - b. What is the turnaround time for each process?
 - c. What is the waiting time for each process?
 - d. What is the CPU utilization rate?
4. The following processes are being scheduled using a preemptive, priority-based, round-robin scheduling algorithm.

Process	Priority	Burst	Arrival
P ₁	8	15	0
P ₂	3	20	0
P ₃	4	20	20
P ₄	4	20	25
P ₅	5	5	45
P ₆	5	15	55

Each process is assigned a numerical priority, with a higher number indicating a higher relative priority. The scheduler will execute the highest-priority process. For processes with the same priority, a round-robin scheduler will be used with a time quantum of 10 units. If a process is preempted by a higher-priority process, the preempted process is placed at the end of the queue.

- Show the scheduling order of the processes using a Gantt chart.
- What is the turnaround time for each process?
- What is the waiting time for each process?